ECE 340: Semiconductor Devices

Lecture 2: Intro to Semiconductor Electronics & Crystal Structure

Prof. Hyunseok Kim

Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

E-mail: hyunseok@illinois.edu

Office Hours: Thursdays 12p-1p, 2256 HMNTL

Important Information

- Download and Review Syllabus / Course Information from Website
- Canvas: Homework Submission, Exams, & Grades
- Course Director: Prof. John Dallesasse
  - jdallesa@illinois.edu
  - Coordinates schedule, policies, absence issues, homework, quizzes, exams, etc.
- Contact Information and Office Hours in Syllabus
- Lecture Slides: Previous Semester Posted on Website
- Recordings of Sp 21 lectures by Prof. Dallesasse on website
- DRES Students: Contact Prof. Dallesasse ASAP
- Textbook:
  - "Solid State Electronic Devices," Streetman & Banerjee, 7th Edition
  - Supplemental: "Semiconductor Device Fundamentals," Pierret
  - Additional reference texts listed in syllabus

Instructional Objectives (1)

By the time of exam No. 1 (after 17 lectures), the students should be able to do the following:

1. Outline the classification of solids as metals, semiconductors, and insulators and distinguish direct and indirect semiconductors.
2. Determine relative magnitudes of the effective mass of electrons and holes from an E(k) diagram.
3. Calculate the carrier concentration in intrinsic semiconductors.
4. Apply the Fermi-Dirac distribution function to determine the occupation of electron and hole states in a semiconductor.
5. Calculate the electron and hole concentrations if the Fermi level is given; determine the Fermi level in a semiconductor if the carrier concentration is given.
6. Determine the variation of electron and hole mobility in a semiconductor with temperature, impurity concentration, and electrical field.
7. Apply the concept of compensation and space charge neutrality to calculate the electron and hole concentrations in compensated semiconductor samples.
8. Determine the current density and resistivity from given carrier densities and mobilities.
9. Calculate the recombination characteristics and excess carrier concentrations as a function of time for both low level and high level injection conditions in a semiconductor.
10. Use quasi-Fermi levels to calculate the non-equilibrium concentrations of electrons and holes in a semiconductor under uniform photoexcitation.
11. Calculate the drift and diffusion components of electron and hole currents.
12. Calculate the diffusion coefficients from given values of carrier mobility through the Einstein's relationship and determine the built-in field in a non-uniformly doped sample.

https://my.ece.illinois.edu/courses/description.asp?ECE340

Instructional Objectives (2)

By the time of Exam No.2 (after 32 lectures), the students should be able to do all of the items listed under A, plus the following:

13. Calculate the contact potential of a p-n junction.
14. Estimate the actual carrier concentration in the depletion region of a p-n junction in equilibrium.
15. Calculate the maximum electrical field in a p-n junction in equilibrium.
16. Distinguish between the current conduction mechanisms in forward and reverse biased diodes.
17. Calculate the minority and majority carrier currents in a forward or reverse biased p-n junction diode.
18. Predict the breakdown voltage of a p+-n junction and distinguish whether it is due to avalanche breakdown or Zener tunneling.
19. Calculate the charge storage delay time in switching p-n junction diodes.
20. Calculate the capacitance of a reverse biased p-n junction diode.
21. Calculate the capacitance of a forward biased p-n junction diode.
22. Predict whether a metal-semiconductor contact will be a rectifying contact or an ohmic contact based on the metal work function and the semiconductor electron affinity and doping.
23. Calculate the electrical field and potential drop across the neutral regions of wide base, forward biased p+-n junction diode.
24. Calculate the voltage drop across the quasi-neutral base of a forward biased narrow base p+-n junction diode.
25. Calculate the excess carrier concentrations at the boundaries between the space-charge region and the neutral n- and p-type regions of a p-n junction for either forward or reverse bias.

Instructional Objectives (3)

By the time of the Final Exam, after 44 class periods, the students should be able to do all of the items listed under A and B, plus the following:

26. Calculate the terminal parameters of a BJT in terms of the material properties and device structure.

27. Estimate the base transport factor “B” of a BJT and rank-order the internal currents which limit the gain of the transistor.

28. Determine the rank order of the electrical fields in the different regions of a BJT in forward active bias.

29. Calculate the threshold voltage of an ideal MOS capacitor.

30. Predict the C-V characteristics of an MOS capacitor.

31. Calculate the inversion charge in an MOS capacitor as a function of gate and drain bias voltage.

32. Estimate the drain current of an MOS transistor above threshold for low drain voltage.

33. Estimate the drain current of an MOS transistor at pinch-off.

34. Distinguish whether a MOSFET with a particular structure will operate as an enhancement or depletion mode device.

35. Determine the short-circuit current and open-circuit voltage for an illuminated p/n junction solar cell.

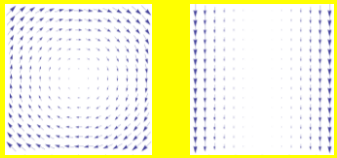
General Introduction to Semiconductor Electronics

Basic Equations for Semiconductor Device Operation

- The basic equations for semiconductor device operation describe the static and dynamic behavior of carriers in semiconductors under the influence of external fields that cause deviation from the thermal equilibrium conditions
- The basic equations can be classified in four groups:
  - Maxwell's Equations
  - Current-Density Equations
  - Continuity Equations
  - Quantum Mechanics & Schrödinger's Equation

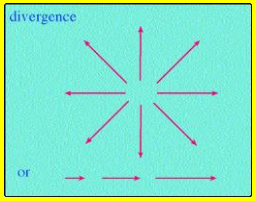
Maxwell Equations for Homogeneous and Isotropic Materials

$$\nabla \times E = -\frac{\partial B}{\partial t}$$



$$\nabla \times H = -\frac{\partial D}{\partial t} + J_{cond} = J_{total}$$

$$\nabla \cdot D = \rho(x,y,z)$$



$$\nabla \cdot B = 0$$

$$B = \mu_o H$$

$$D(r,t) = \int_{-\infty}^{t} \varepsilon_s(t-t')E(r,t')dt'$$

$$= \varepsilon E + P$$

E: electric field

D: electric displacement

B: magnetic field

H: magnetizing field

ε_s: permittivity

μ_o: permeability

ρ: total electric charge density

J_{cond}: the conduction current density

P: polarization density

$\times$: curl operator

$\bullet$: divergence operator

Key Points: Fields & Waves

Charge creates divergence in the electric displacement field:

$$\nabla \cdot D = \rho$$

The electric field is related to the electric displacement field through the permittivity:

$$D = \varepsilon \mathcal{E}$$

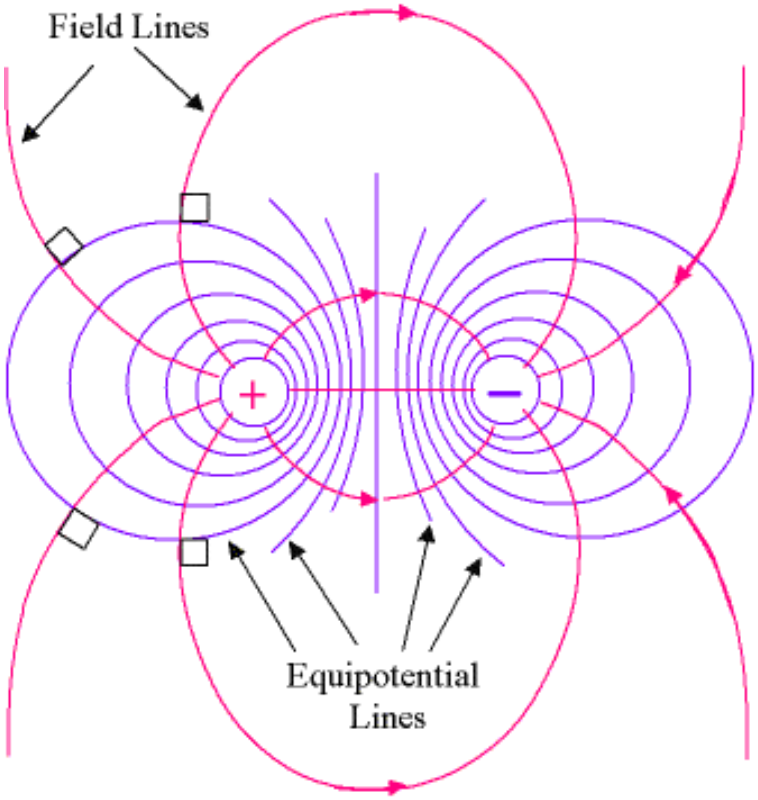
$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}$$

The electric field is the negative of the gradient of the scalar potential field:

$$\mathcal{E} = -\nabla \mathcal{V}$$

In one dimesion: $\mathcal{E}(x) = -\dfrac{d\mathcal{V}(x)}{dx}$

$$\mathcal{V}(x) = -\int \mathcal{E}(x)\,dx$$



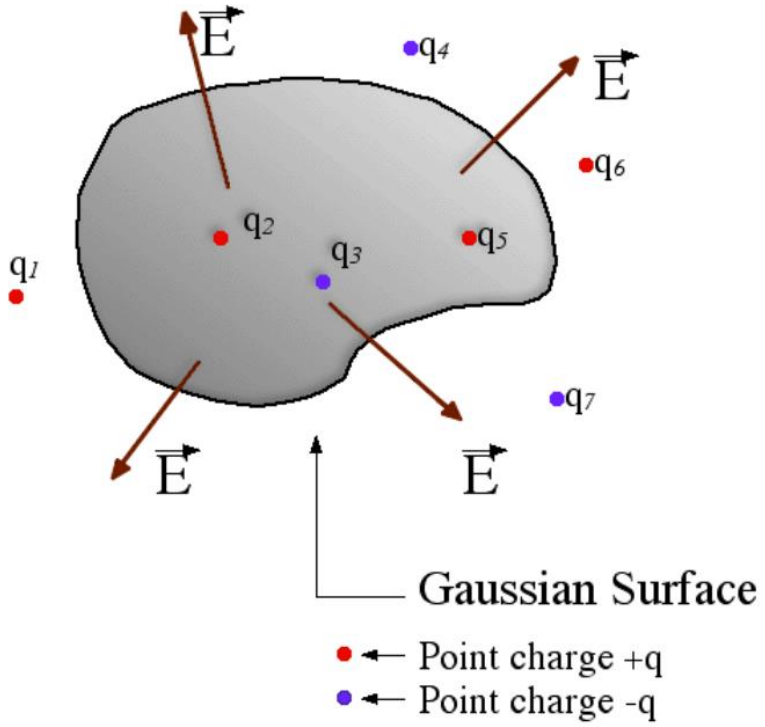
http://www.pstcc.edu/departments/natural_behavioral_sciences/Web%20Physics/E2020D0103.gif

Gauss' Law

$$\nabla \cdot \mathbf{D} = \rho$$

ρ is the charge density with units of $\frac{Coulombs}{m^3}$

- The divergence of the electric displacement flux density is equal to the charge density



http://www.ibiblio.org/links/devmodules/Gauss/graphics/blob1.gif

Key Points: Circuit Concepts

Capacitance :

Simple Form: $C = \left|\frac{Q}{V}\right|$

Differential Form: $C = \left|\frac{dQ}{dV}\right|$

Resistance / Conductance :

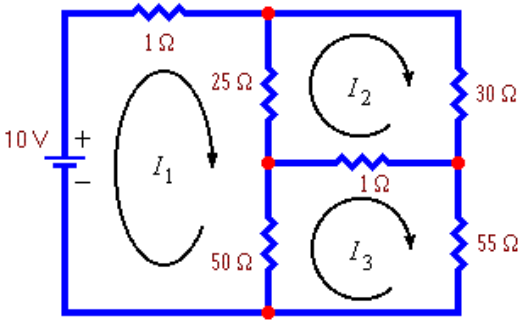
Ohm's Law: $V = IR \quad (I = JA)$

Resistance (AC): $R = \frac{dV}{dI}$

Conductance (AC): $G = \frac{dI}{dV}$

Bulk Resistivity: $\rho = R\frac{A}{L} = \frac{1}{\sigma}$

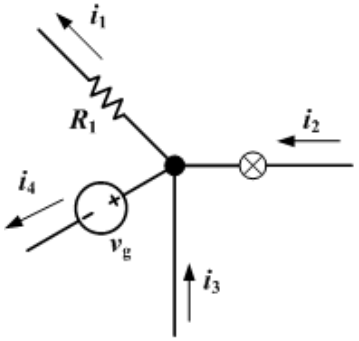
Loop Analysis



$$\begin{cases} 1i_1 + 25(i_1 - i_2) + 50(i_1 - i_3) - 10 = 0 \\ 25(i_2 - i_1) + 30i_2 + 1(i_2 - i_3) = 0 \\ 50(i_3 - i_1) + 1(i_3 - i_2) + 55i_3 = 0 \end{cases}$$

http://mathonweb.com/help/backgd4.htm

Node Analysis



$$i_1 + i_4 = i_2 + i_3$$

Current density Equations

$$J = qnv$$

q: charge per carrier
n: carrier concentration
v: average carrier velocity

Drift Diffusion

$$J_n = q\mu_n nE + qD_n \nabla n$$

$$J_p = q\mu_p pE - qD_p \nabla p$$

$$J_{cond} = J_n + J_p$$

We will revisit this later

- In semiconductor, in addition to an **electron** current density there is a **hole** current density
- Each current consist of the **drift** component cause by field and the **diffusion** component caused by the carrier concentration gradient

Continuity Equations

$$\frac{\partial n}{\partial t} = G_n - U_n + \frac{1}{q}\nabla \bullet J_n$$

$$\frac{\partial p}{\partial t} = G_p - U_p - \frac{1}{q}\nabla \bullet J_p$$

We will revisit this later

For a given volume of semiconductor, the rate change of carrier is the net effect of current flow into the volume and generation and recombination rates within the volume.

G_n : *electron generation rate*

G_p : *hole generation rate*

U_n : *electron recombination rate*

U_p : *hole recombination rate*

Quantum Mechanics

Heisenberg Uncertainty Principle:

Position-Momentum: $(\Delta x\, \Delta p_x) \geq \frac{\hbar}{2}$

Energy-Time: $(\Delta E\, \Delta t) \geq \frac{\hbar}{2}$

Wave function: Ψ(x,y,z,t)

- Describes the quantum state of a particle
- Ψ itself is generally complex and is not directly observable
- The probability density of finding the particle at (x,y,z) is $|\Psi|^2 = \Psi^* \Psi$
- The particle is somewhere, so $\int_{-\infty}^{\infty} \Psi^* \Psi \, dx\, dy\, dz = 1$

Schrödinger's equation

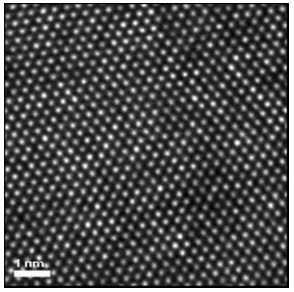
$$-\frac{\hbar^2}{2m}\nabla^2\Psi + V\Psi = i\hbar\frac{\partial\Psi}{\partial t}$$

(kinetic energy + potential energy = total energy)

Lattices and Crystal Structures

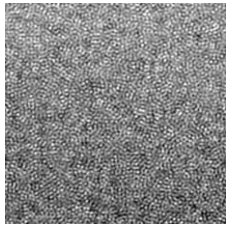
Types of Solids

HRTEM: [110] Si



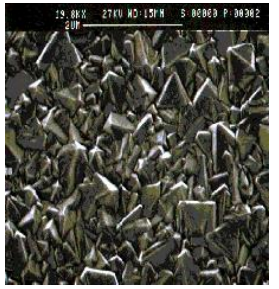
http://spie.org/x31901.xml

TEM: Glassy Pd

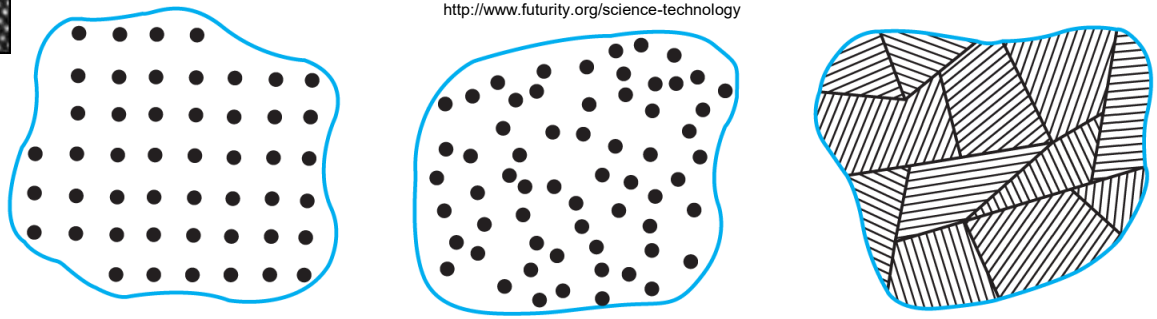


http://www.futurity.org/science-technology

SEM: Diamond on Si



http://isic.epfl.ch/page-53777-en.html



(a) Crystalline (b) Amorphous (c) Polycrystalline

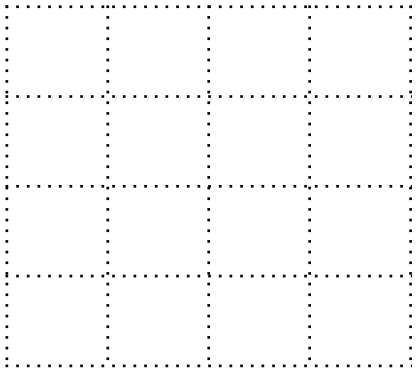
Figure 1.1

Three types of solids, classified according to atomic arrangement: (a) crystalline and (b) amorphous materials are illustrated by microscopic views of the atoms, whereas (c) polycrystalline structure is illustrated by a more macroscopic view of adjacent single-crystalline regions, such as (a).

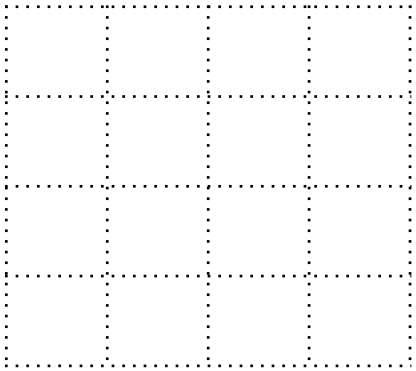
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Crystal Lattice

- **Lattice:** Periodic arrangement of ‘lattice points’
- **Basis:** atom or group of atoms attached identically to every lattice point.
- Repeating the basis at every lattice point constructs the crystal structure.



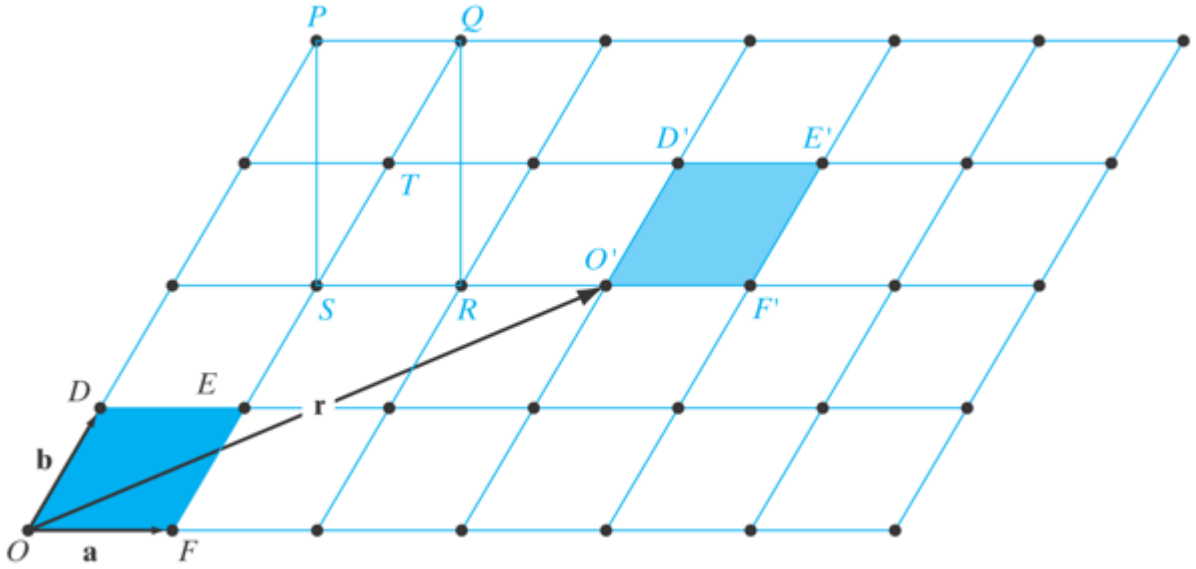
+ = ?



+ = ?

Crystal Lattice

- **Unit Cell:** Contains a region which is representative of the lattice which can be regularly repeated to recreate the entire lattice
- **Primitive Cell:** The smallest unit cell that can be repeated in integral steps to produce the lattice. Contains a single lattice point
- **Primitive Vectors:** Describe how the lattice repeats in space (smallest translations generating the lattice).
- **Basis Vectors**: Similar to primitive vectors, but used to replicate the lattice through the translation of a specific unit cell



Simple Lattices and Unit Cells

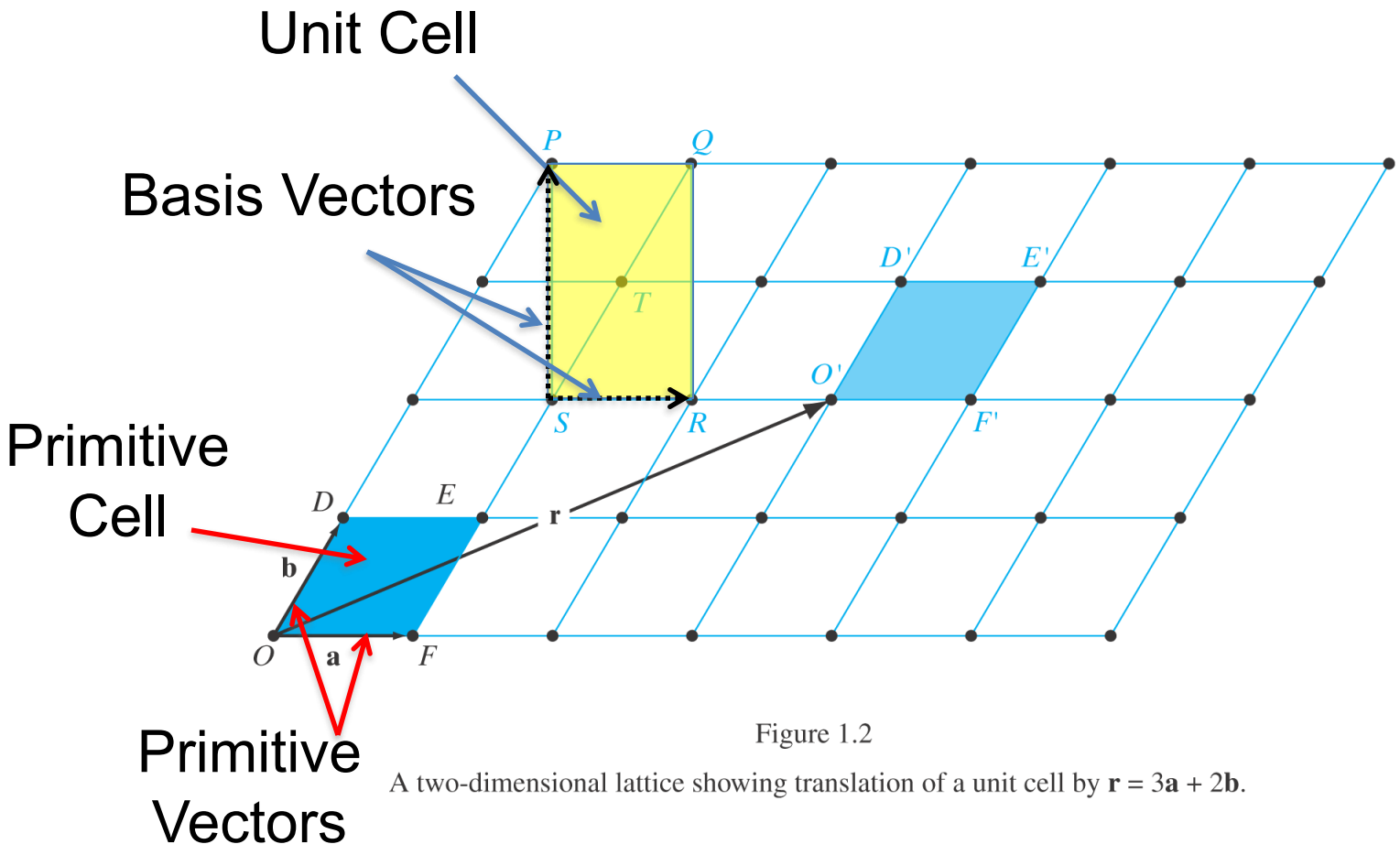


Figure 1.2

A two-dimensional lattice showing translation of a unit cell by $\mathbf{r} = 3\mathbf{a} + 2\mathbf{b}$.

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Other Important Definitions

- **Lattice Constant:** distance along the edge of a unit cell ("a" in examples that follow)
  - More generally, the length of the basis vectors

NOTE: The lattice constant, in general, is **NOT** the distance between atoms (bond length).

Cubic Crystal Structures

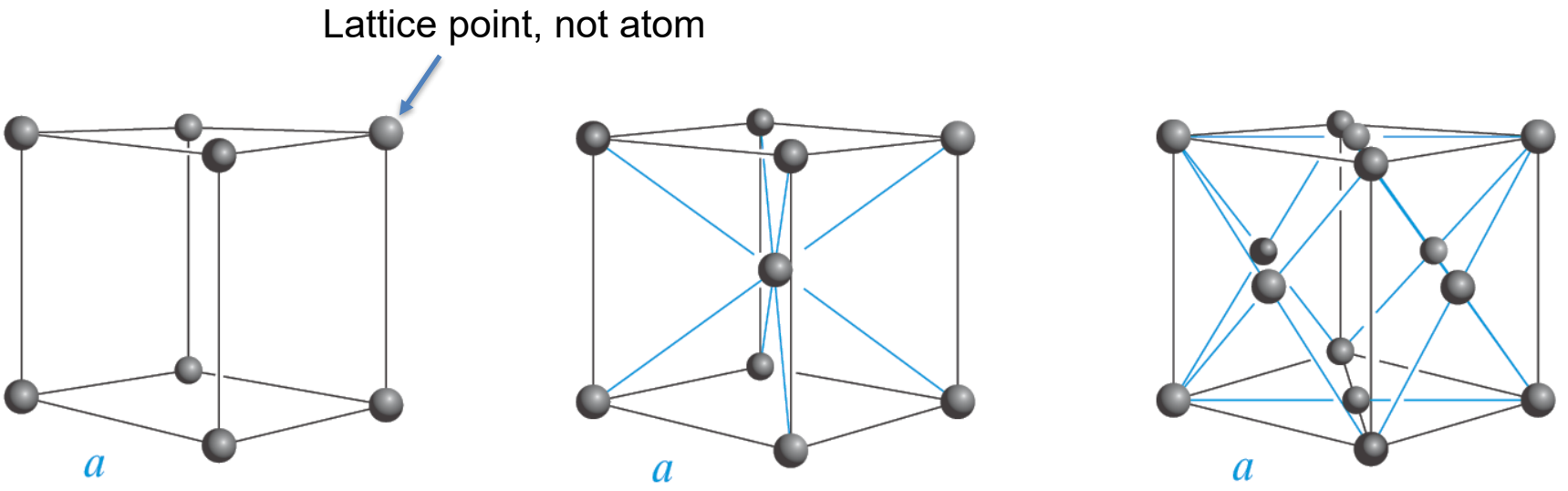


Figure 1.3

Unit cells for three types of cubic lattice structures.

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Cubic Crystal Structures

- **<u>Calculating the number of 'lattice points' in a unit cell:</u>** Count each lattice point according to the fraction belonging to the unit cell.

How many lattice points for below unit cells?

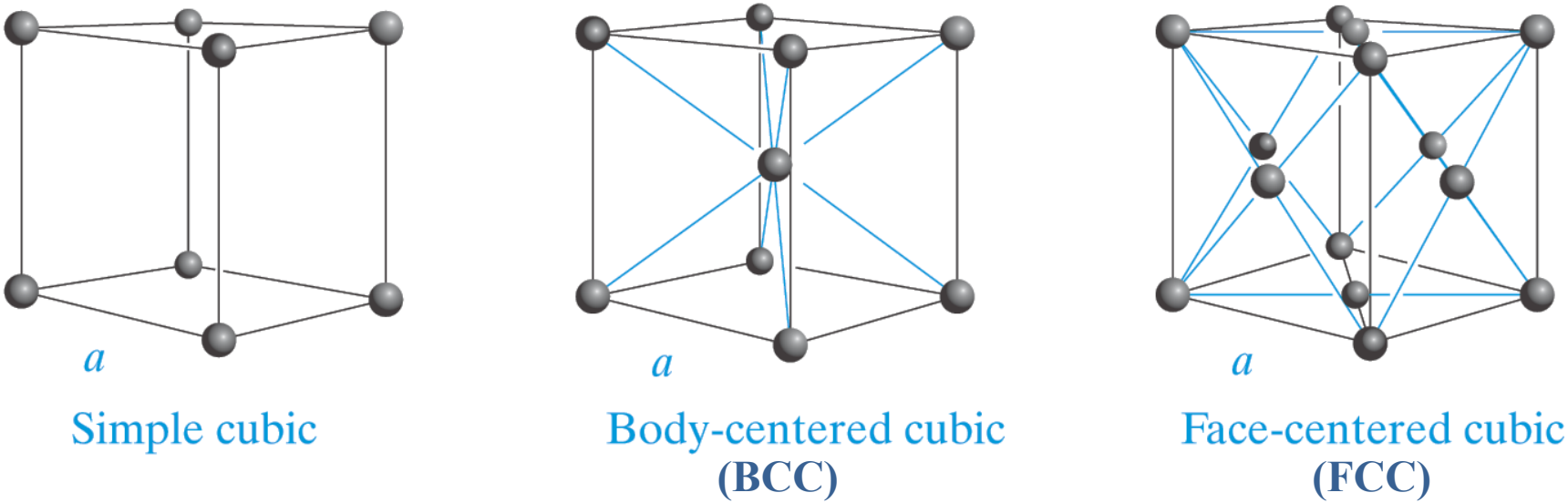


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Primitive Cells

BCC and FCC

Primitive Cell: BCC and FCC

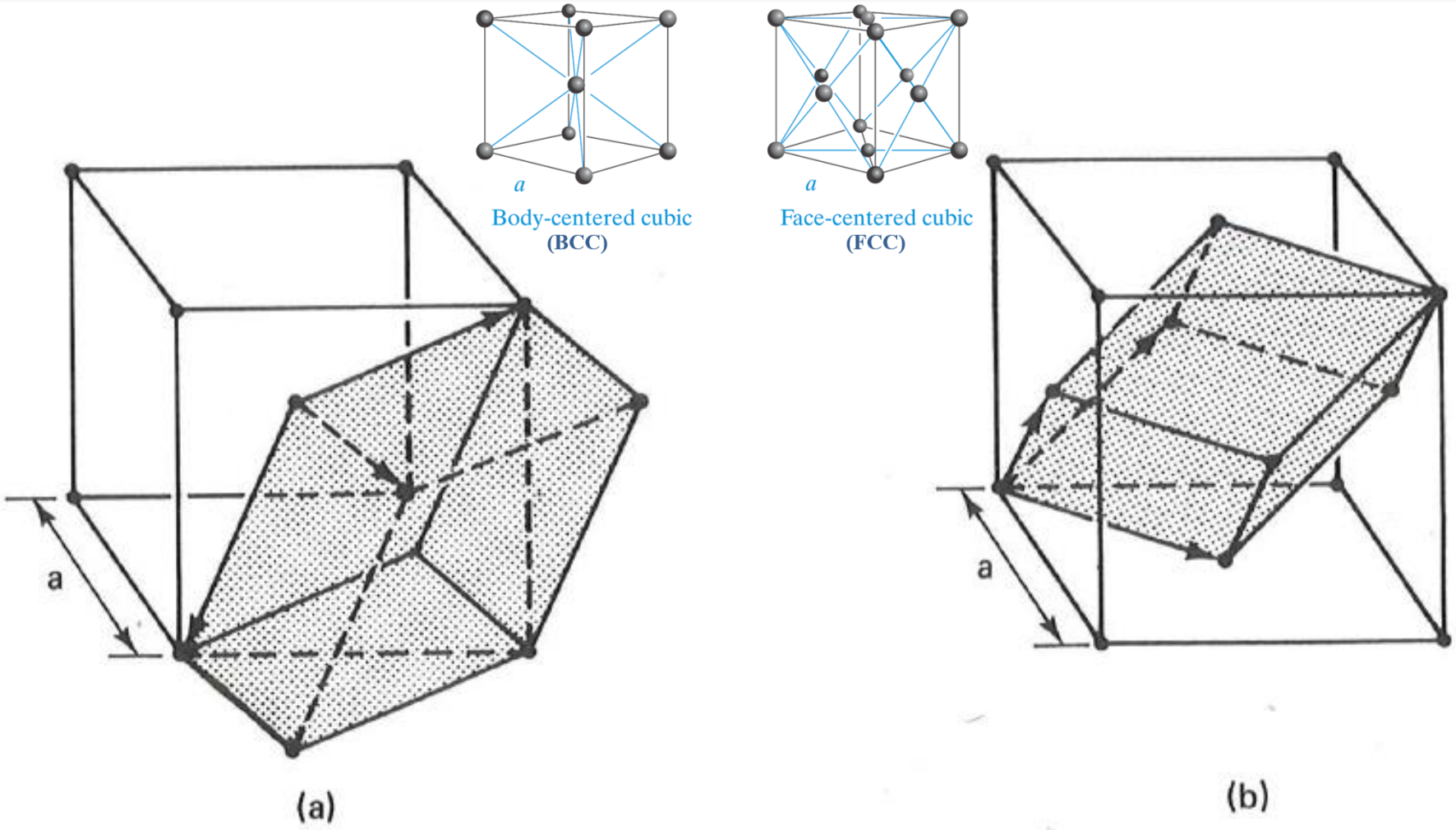
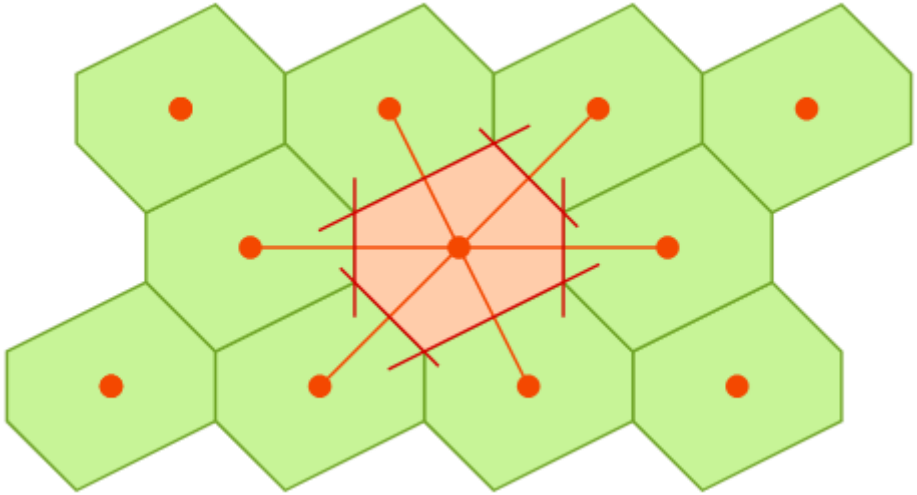


Figure 1.7 Cubic and primitive unit cells for (a) the body-centered and (b) the face-centered cubic Bravais lattices.

From: Wolfe, Holonyak, & Stillman, "Physical Properties of Semiconductors

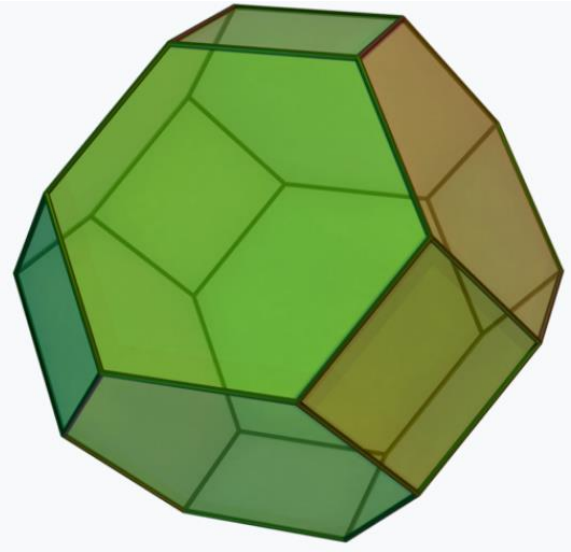
Wigner-Seitz Primitive Cell

- Draw a line from a given lattice point to all nearby lattice points
- Bisect the lines with orthogonal planes
- Construct the smallest polyhedron that contains the given lattice point

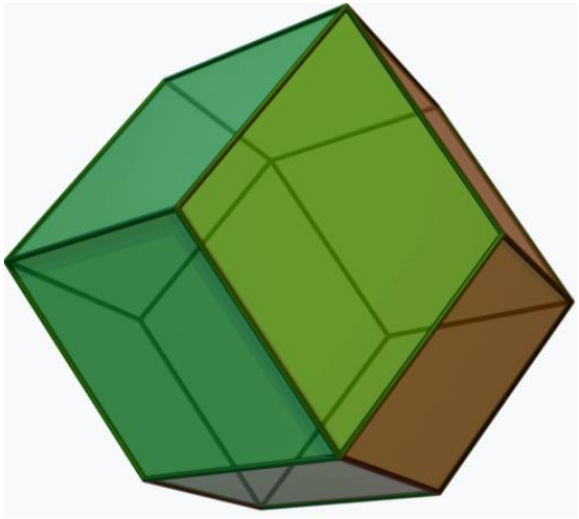


Wigner-Seitz Primitive Cell

BCC: truncated octahedron



FCC: rhombic dodecahedron



Diamond Structure

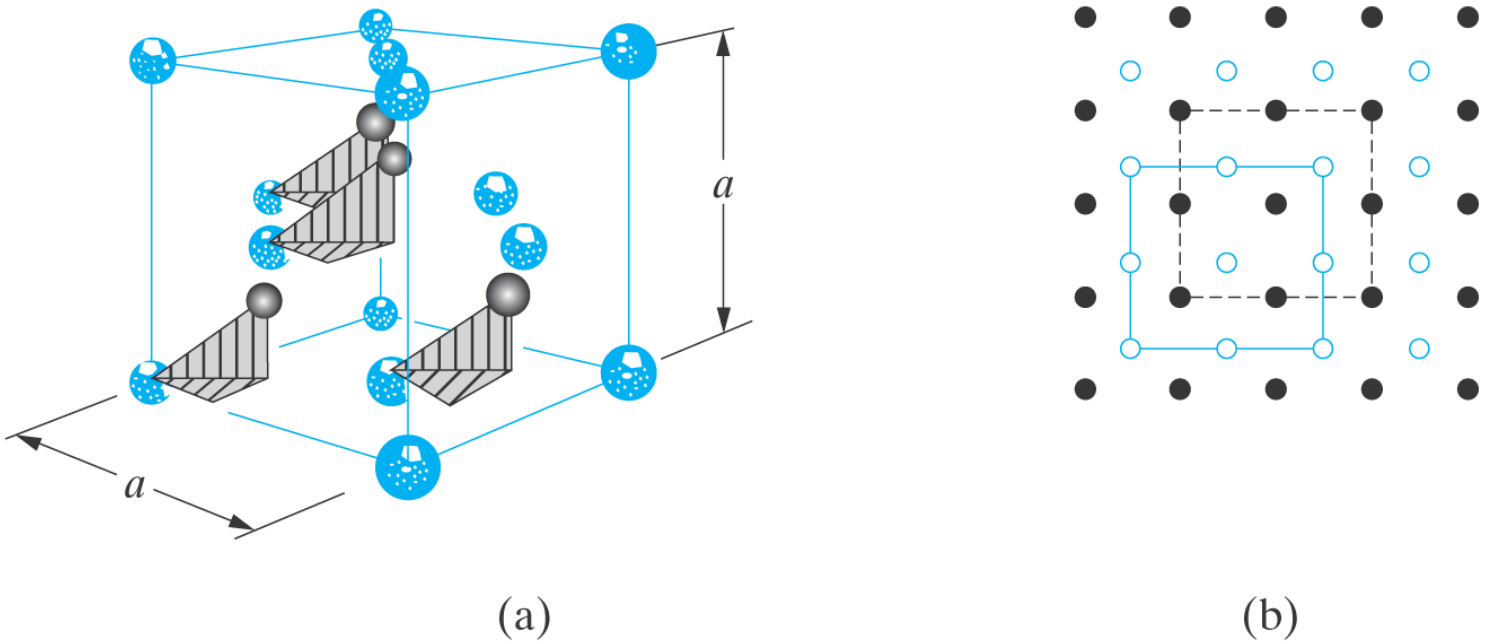


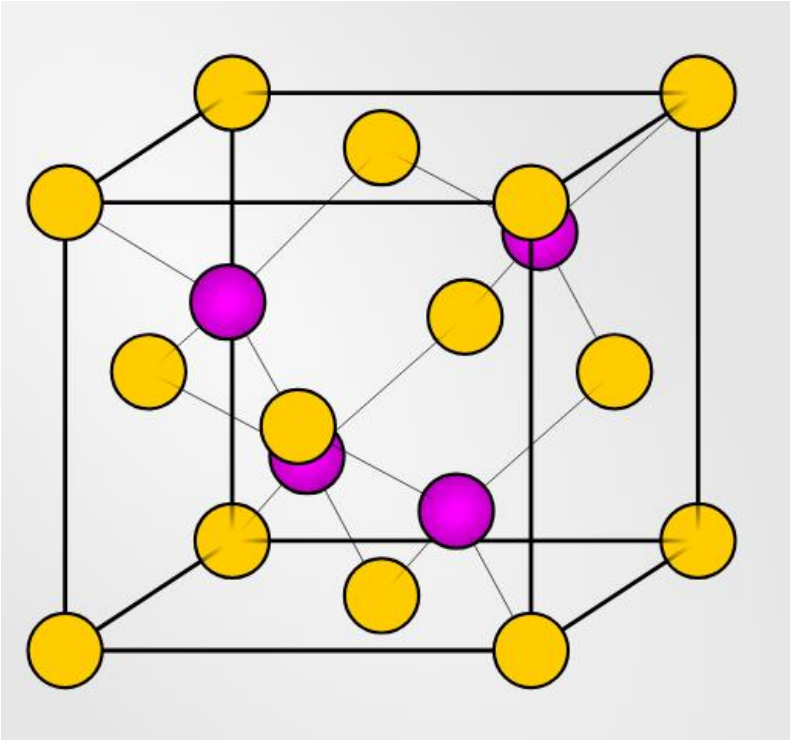
Figure 1.8

Diamond lattice structure: (a) a unit cell of the diamond lattice constructed by placing atoms $\frac{1}{4},\frac{1}{4},\frac{1}{4}$ from each atom in an fcc; (b) top view (along any ⟨100⟩ direction) of an extended diamond lattice. The colored circles indicate one fcc sublattice and the black circles indicate the interpenetrating fcc.

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Diamond Structure Bonding

Two superimposed FCC



Nearest neighbor structure

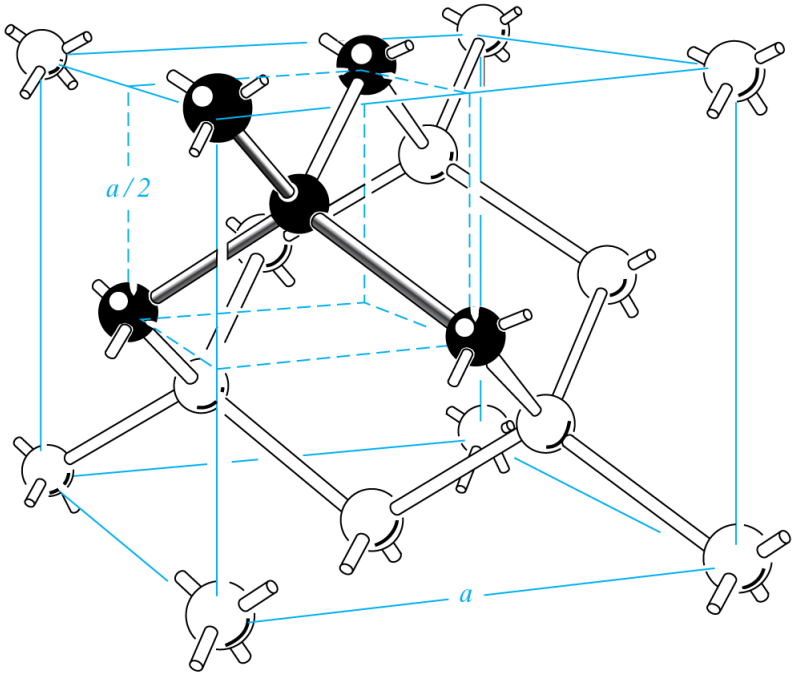


Figure 1.9

Diamond lattice unit cell, showing the four nearest neighbor structure.

(From *Electrons* and *Holes in Semiconductors* by W. Shockley, © 1950 by Litton Educational Publishing Co., Inc.; by permission of Van Nostrand Reinhold Co., Inc.)

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Diamond Structure Bonding

Diamond Unit Cell (not primitive cell)

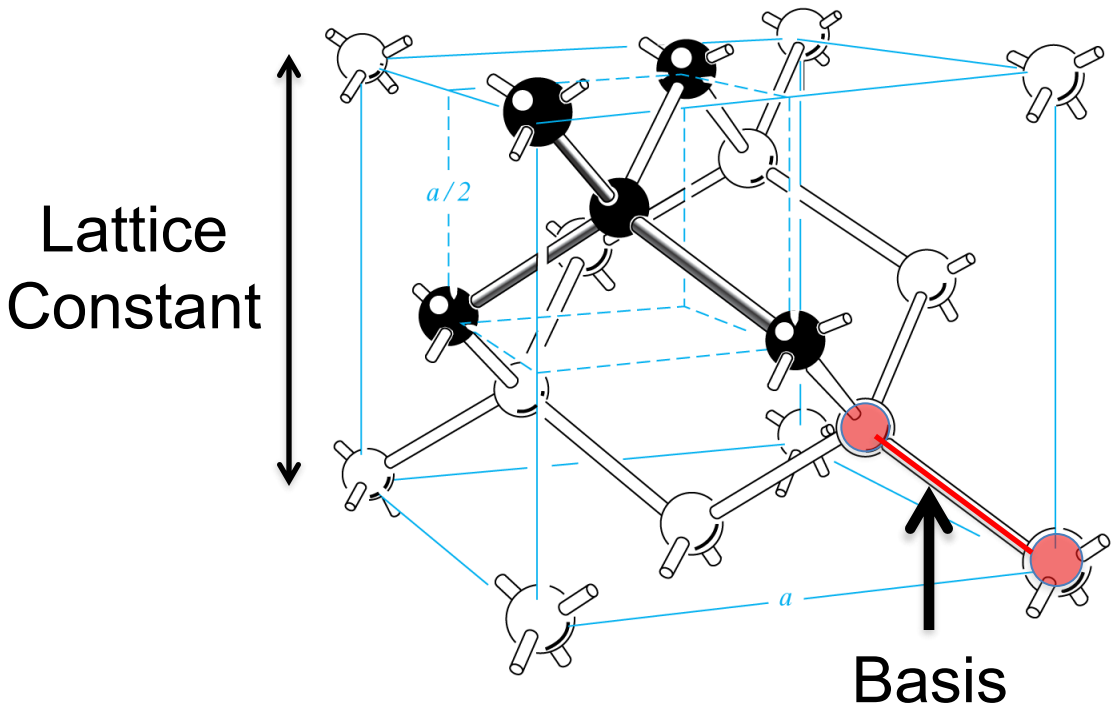


Figure 1.9

Diamond lattice unit cell, showing the four nearest neighbor structure.

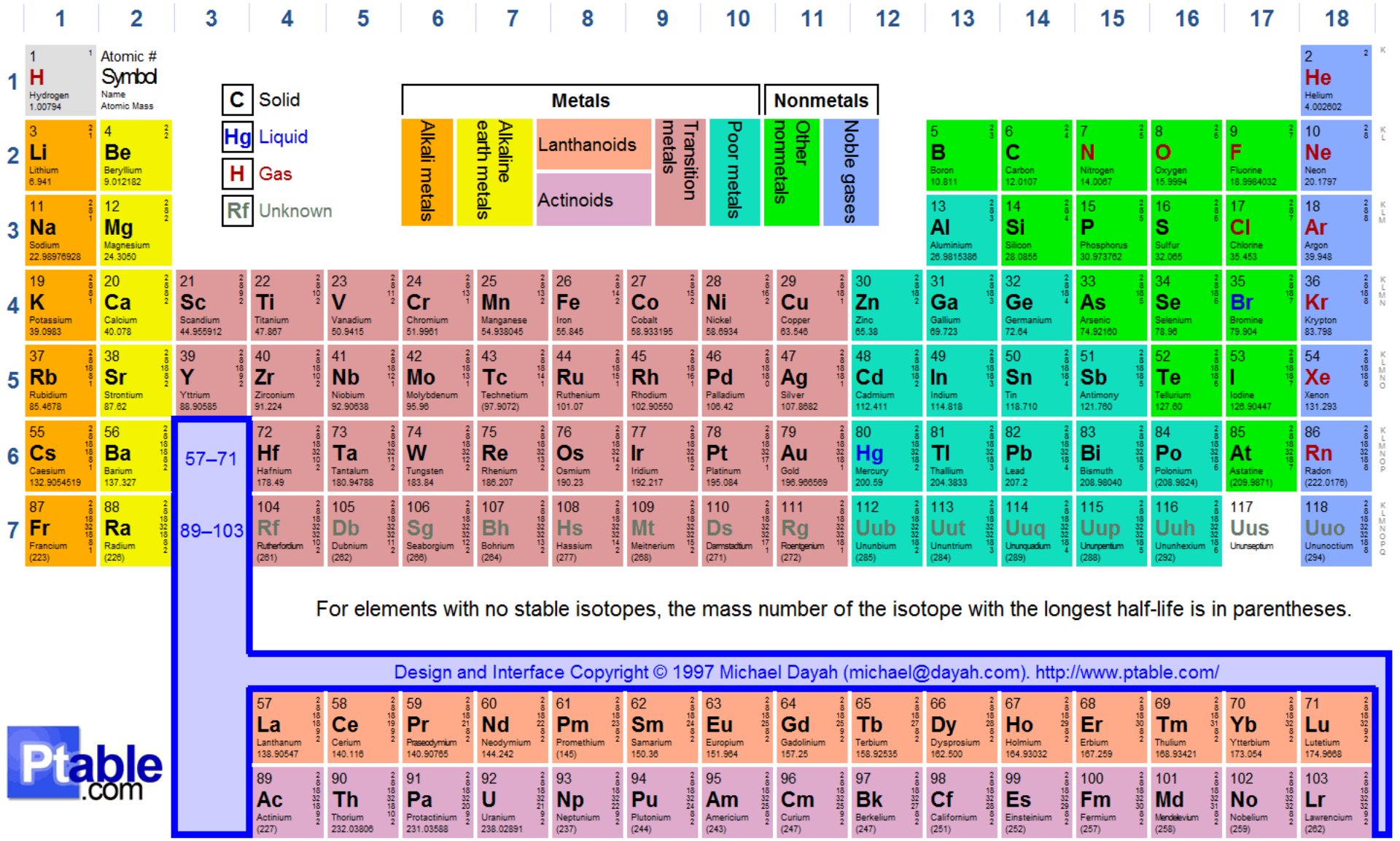
(From *Electrons* and *Holes in Semiconductors* by W. Shockley, © 1950 by Litton Educational Publishing Co., Inc.; by permission of Van Nostrand Reinhold Co., Inc.)

- The **Diamond Structure** is an **FCC Lattice** with a **Basis** of 2 atoms
- The **Unit Cell** has **4 Lattice Points**, and **8 Atoms**

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Semiconductor Materials

Periodic Table of the Elements



For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

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Ptable.com

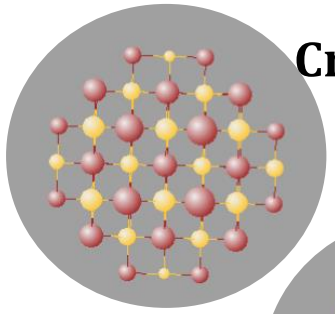
What is a Semiconductor?

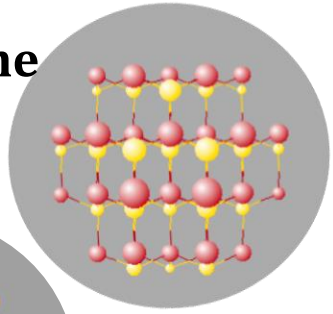
	III	IV	V	
	5 B Boron 10.811	6 C Carbon 12.0107	7 N Nitrogen 14.0067	8 O Oxygen 15.9994
	13 Al Aluminium 26.9815386	14 Si Silicon 28.0855	15 P Phosphorus 30.973762	16 S Sulfur 32.065
30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.64	33 As Arsenic 74.92160	34 Se Selenium 78.96
48 Cd Cadmium 112.411	49 In Indium 114.818	50 Sn Tin 118.710	51 Sb Antimony 121.760	52 Te Tellurium 127.60
80 Hg Mercury 200.59	81 Tl Thallium 204.3833	82 Pb Lead 207.2	83 Bi Bismuth 208.98040	84 Po Polonium (208.9824)

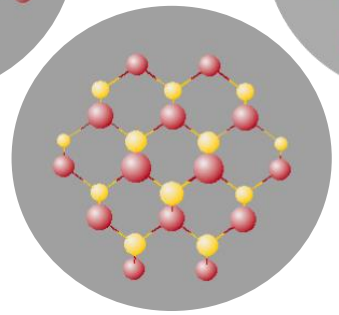


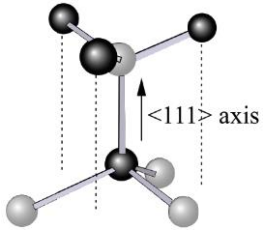


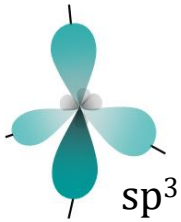
A Crystalline Solid









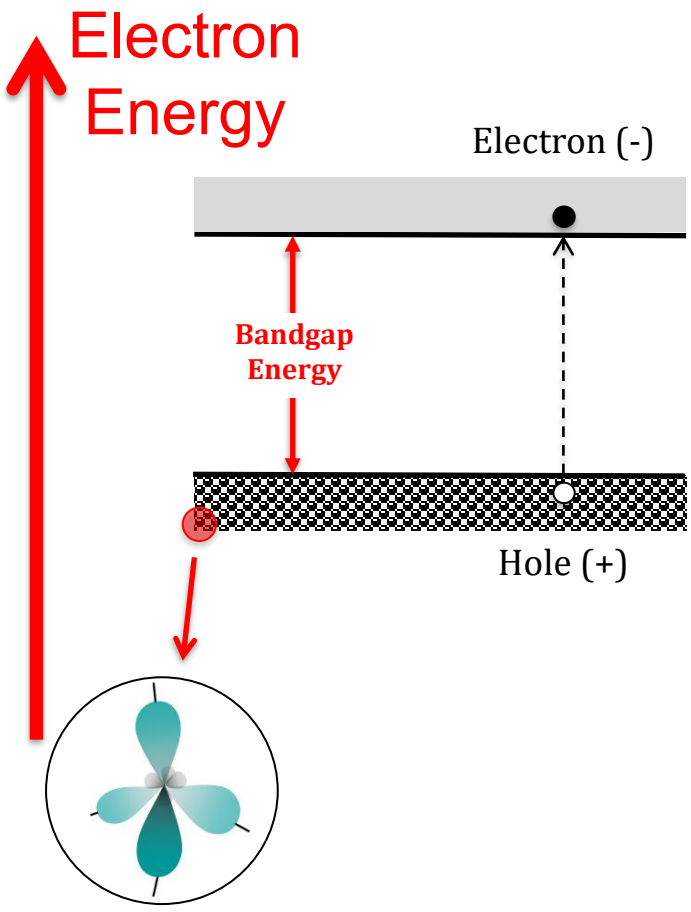


sp^3 Hybridization

Important Material Systems

- Elemental Semiconductors
  - Silicon
    - ICs, CCDs, Solar Panels & Solar Cells
  - Germanium
    - Substrates for High-Efficiency Solar Cells, Long Wavelength Photodetectors
- Binary Semiconductors
  - SiGe
    - High Speed Electronics
  - InP
    - Optical Communication Devices, High Speed Electronics
  - GaAs
    - Wireless Communications, Short-Distance Optical Links
  - GaN
    - LEDs for Lighting, High Temperature Electronics

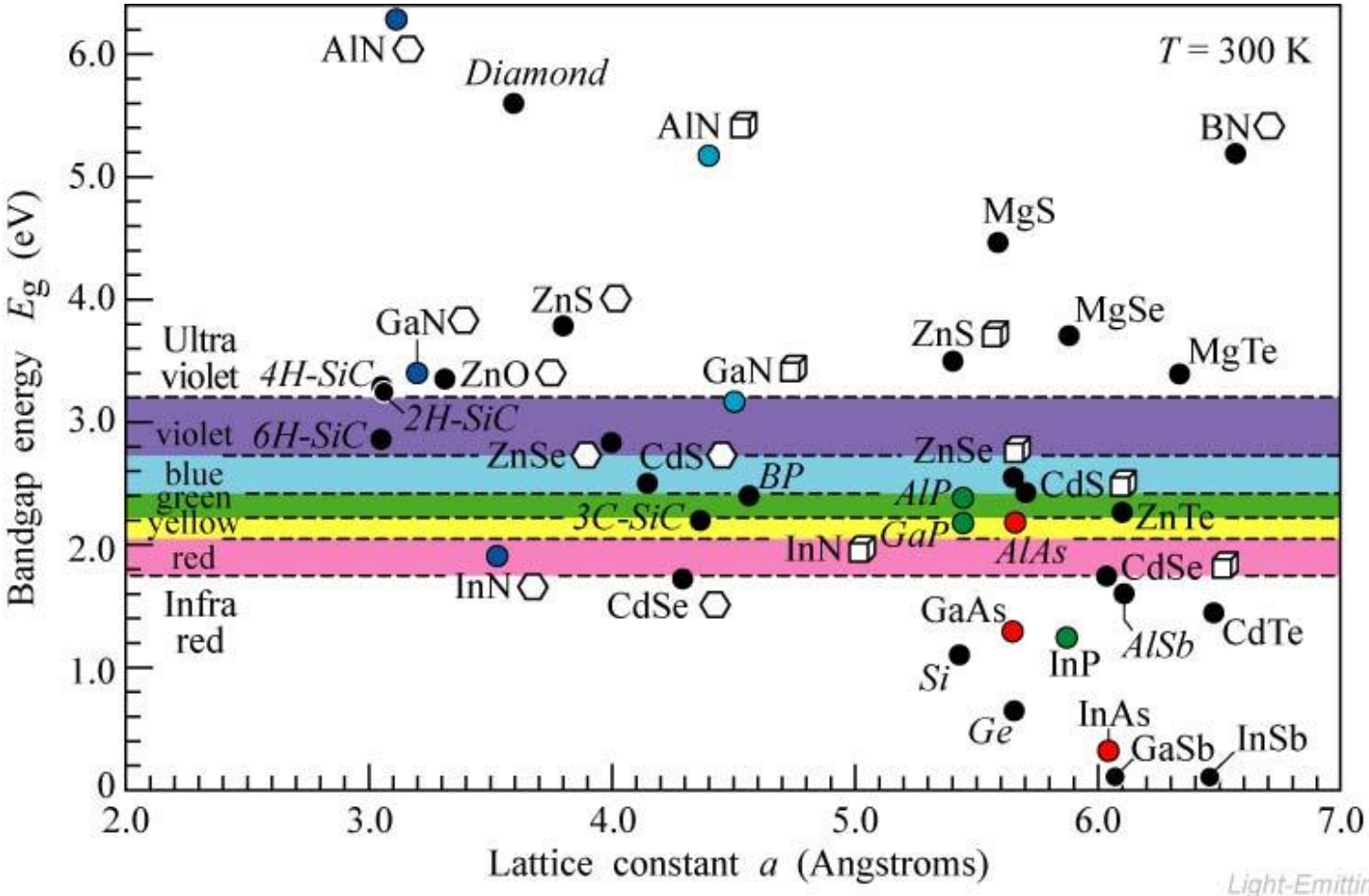
Electrons, Hole, and Energy Bands



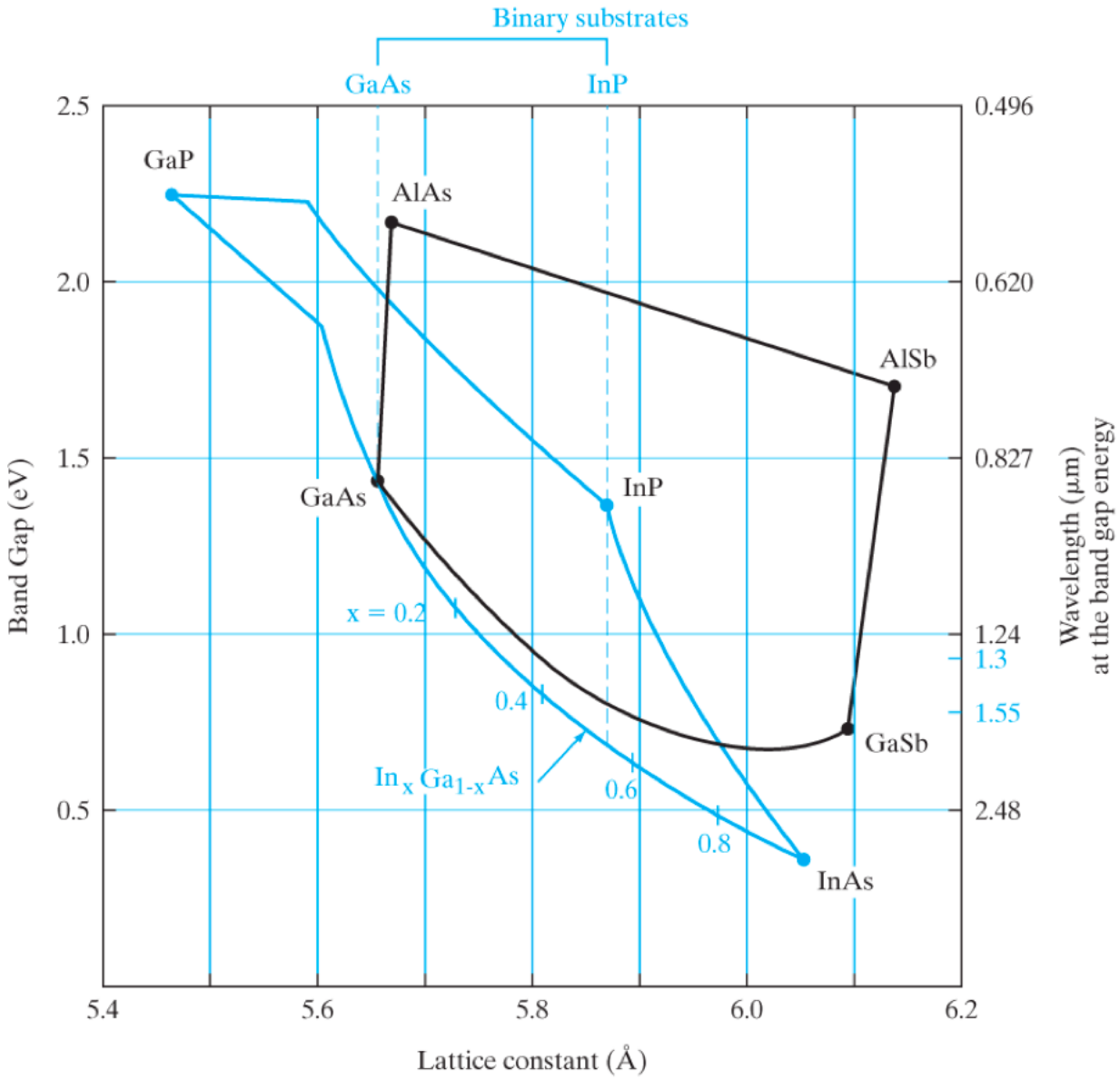
Conduction Band:
Mobile Electrons

Valence Band:
Binding Electrons

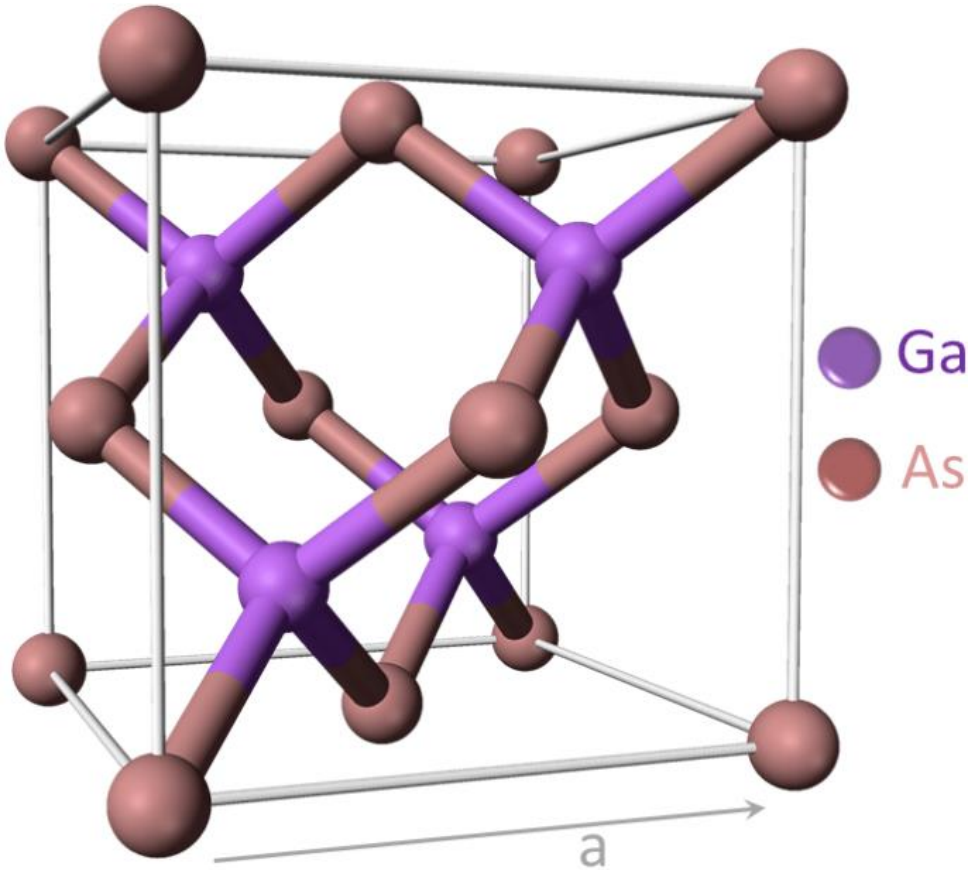
Bandgap and Lattice Constant of Common Semiconductors



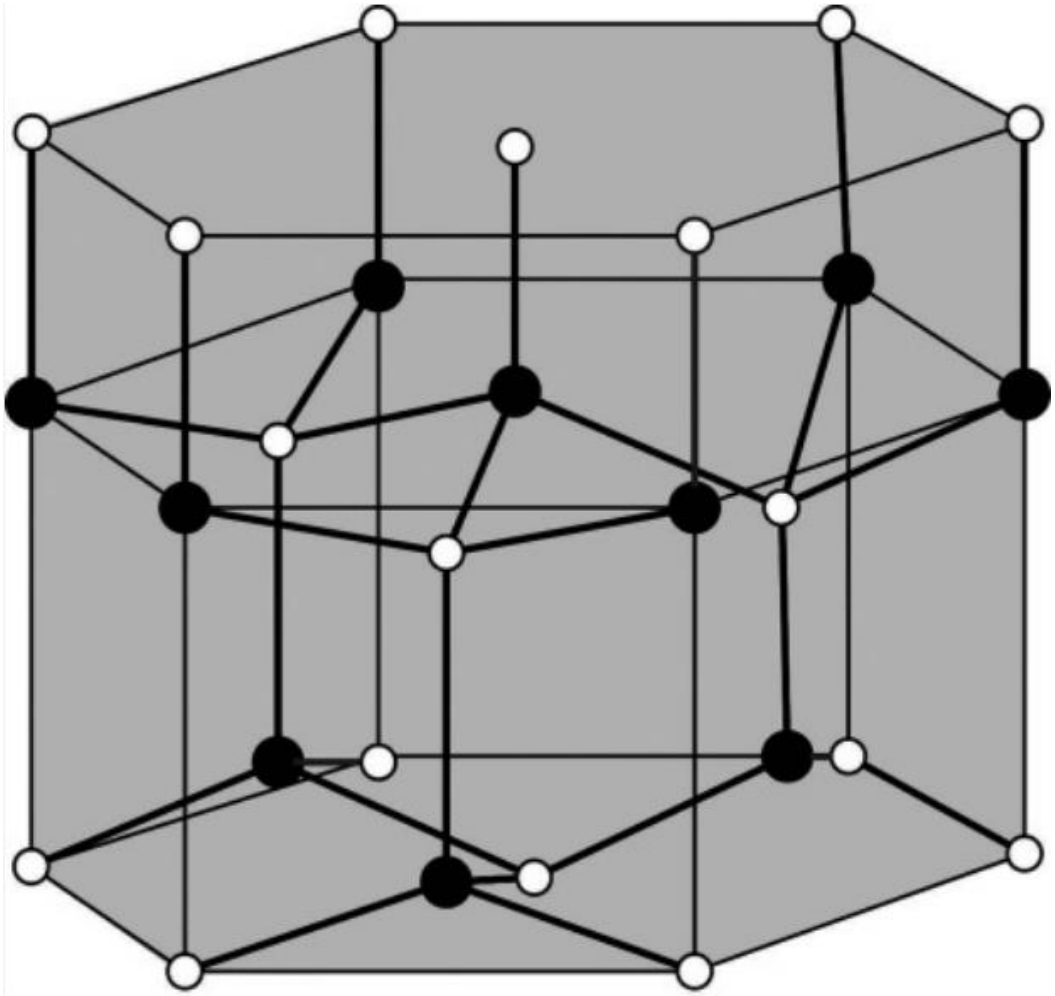
Important Material Systems

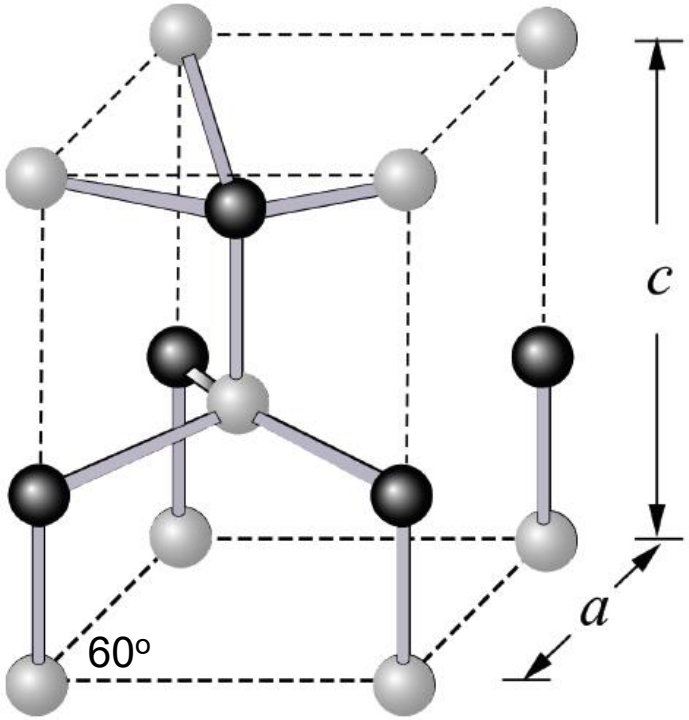


Zinc Blende Structure



Wurtzite Structure





Unit cell of a Wurtzite structure

$$c = a\sqrt{8/3}$$